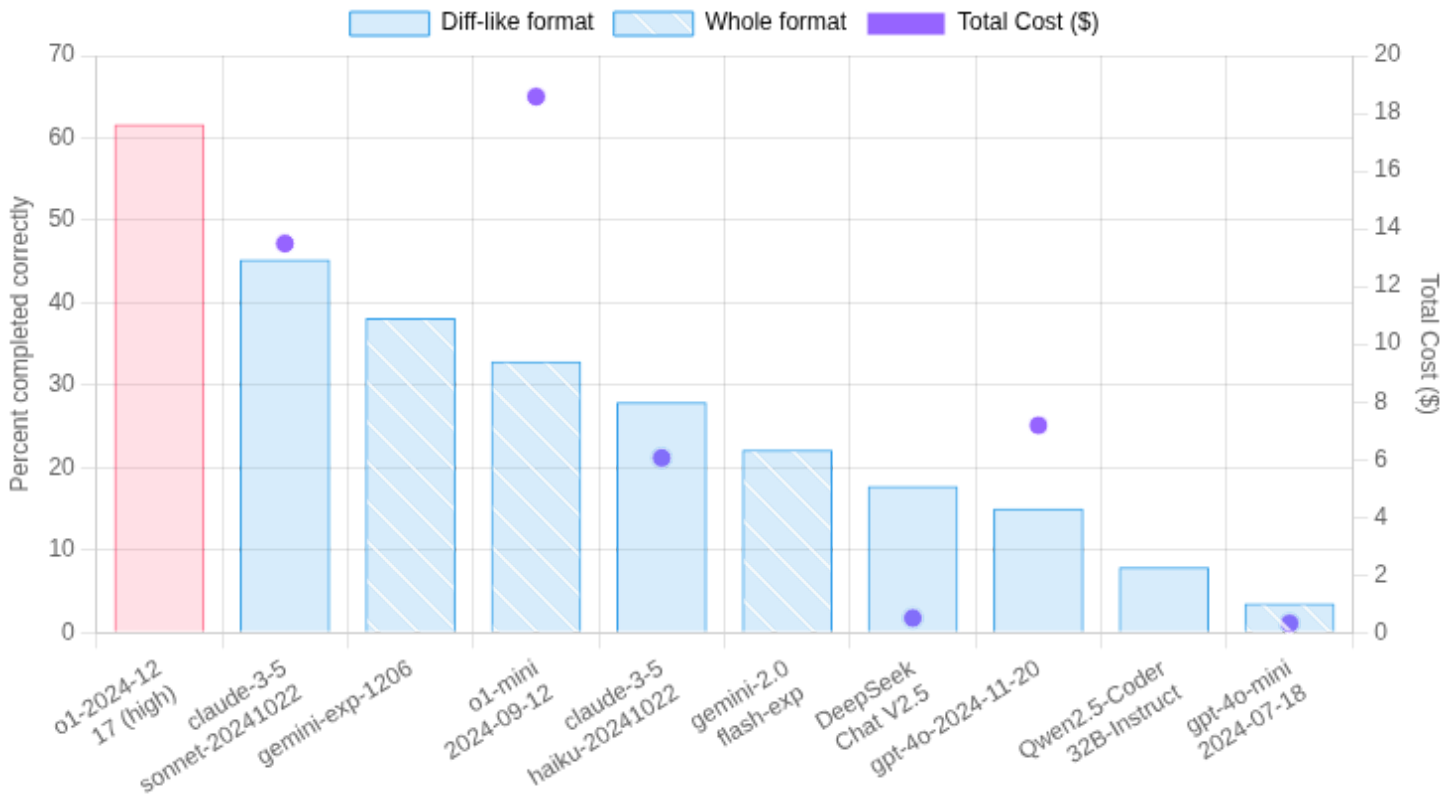
December 21, 2024

o1 tops aider’s new polyglot leaderboard



OpenAI’s new o1 model with “high” reasoning effort gets the top score on the new aider polyglot leaderboard, significantly ahead of other top LLMs. The new polyglot benchmark uses many popular coding languages and was designed to be *much more challenging* than aider’s original code editing benchmark. This more clearly distinguishes the performance of today’s strongest coding models and leaves headroom for future LLMs.

> **NOTE**
> See the main aider leaderboard for benchmark results from more models. This article only contains a snapshot of results at the time of publication.

The polyglot benchmark

Like aider’s original code editing benchmark, the new polyglot benchmark is based on Exercism coding exercises.

The new polyglot benchmark:

- Contains coding problems in C++, Go, Java, JavaScript, Python and Rust. The old benchmark was solely based on Python exercises.
- Focuses on the *most difficult* 225 exercises out of the 697 that Exercism provides for those languages. The old benchmark simply included all 133 Python exercises, regardless of difficulty.

Motivation and goals

Aider’s original code editing benchmark was saturating as the top scores approached and then surpassed 80%. Sonnet’s score of 84.2% was based on solving 112 of the 133 exercises, leaving only 21 unsolved exercises. New champions were advancing the top score by solving just 1-2 more problems than the previous record. This made it hard to clearly measure the difference in code editing skill between these top models.

Part of the problem is that many of the original 133 Python problems are very easy and provide little challenge to today’s frontier LLMs. Models as old as GPT 3.5 Turbo were able to solve half of the 133 problems. Such easy problems simply inflate the benchmark scores of modern LLMs without providing any data about which models are better or worse.

The main goal for a new benchmark was to re-calibrate the scale so that today’s top coding LLMs would occupy a wide range of scores between about 5% and 50%. This should leave headroom for future LLMs and make it possible to more clearly compare the relative performance of top models.

Designing the polyglot benchmark

The new benchmark:

- Tests LLMs with more coding languages, to increase diversity and source a larger pool of problems.
- Includes just the most challenging coding problems and excludes easy problems that are solvable by most of today’s top coding LLMs.
- Includes more total coding problems, to enable more granularity of comparison.

The new benchmark is based on Exercism coding problems from 6 of the most popular programming languages:

- C++

- Go
- Java
- JavaScript
- Python
- Rust

Exercism provides a total of 697 coding problems in those 6 languages. A set of 7 of today’s top coding models each attempted all 697 of the Exercism problems:

- Sonnet
- Haiku
- o1 Mini
- DeepSeek
- GPT-4o
- Qwen 32B Coder Instruct
- GPT-4o Mini

Depending on the difficulty of the problems, a different number of solutions were found by the collection of 7 models:

Solutions found	Number of problems	Cumulative number of problems
0	66	66
1	61	127
2	50	177
3	48	225
4	53	278
5	71	349
6	90	439
7	258	697

In the table above, you can see that 258 of the problems were solved by all 7 LLMs. These problems are far too easy, and wouldn’t be good choices for the new benchmark. Instead, we

need hard problems like the 66 that none of the 7 models were able to solve.

The new benchmark uses the 225 problems that were solved by 3 or fewer models. This achieves a balance between hard and moderate problems, and provides a large but not excessive total pool of problems. It also represents a good diversity of coding languages:

Language	Problems
C++	26
Go	39
Java	47
JavaScript	49
Python	34
Rust	30
Total	**225**

o1

OpenAI's new o1 model established a very strong top score of 62% on the new benchmark. This still leaves 86 problems of headroom for future models to solve. Given the incredible pace of recent advancements, it will be interesting to see how long it will take for this new benchmark to saturate.

Benchmark problems

The 225 coding problems are available in the aider polyglot benchmark repo on GitHub.

Results

Model	Percent completed correctly	Percent using correct edit format	Command	Edit format
o1-2024-12-17 (high)	61.7%	91.5%	`aider --model openrouter/openai/o1`	diff
claude-3-5-sonnet-20241022	45.3%	100.0%	`aider --model claude-3-5-sonnet-20241022`	diff
gemini-exp-1206	38.2%	98.2%	`aider --model gemini/gemini-exp-1206`	whole
o1-mini-2024-09-12	32.9%	96.9%	`aider --model o1-mini`	whole
claude-3-5-haiku-20241022	28.0%	91.1%	`aider --model claude-3-5-haiku-20241022`	diff
gemini-2.0-flash-exp	22.2%	100.0%	`aider --model gemini/gemini-2.0-flash-exp`	whole
DeepSeek Chat V2.5	17.8%	92.9%	`aider --model deepseek/deepseek-chat`	diff
gpt-4o-2024-11-20	15.1%	96.0%	`aider --model gpt-4o-2024-11-20`	diff
Qwen2.5-Coder-32B-Instruct	8.0%	71.6%	`aider --model openai/Qwen/Qwen2.5-Coder-32B-Instruct # via hyperbolic`	diff
gpt-4o-mini-2024-07-18	3.6%	100.0%	`aider --model gpt-4o-mini-2024-07-18`	whole